

Supplementary Material for InTraScale: Accelerating Video Generation via In-Trajectory Resolution Scaling

Anonymous submission

Scope and Evidence Conventions

This supplement provides implementation details, protocol definitions, additional results, and paired statistical analyses for InTraScale. The main paper remains self-contained. All comparisons within a suite use matched text prompts, generation seeds, initial noise, scheduler, guidance, frame count, and output resolution.

Metric naming. We evaluate Subject Consistency, Background Consistency, Motion Smoothness, Aesthetic Quality, and Imaging Quality using the VBench custom-input protocol. We call their unweighted mean *VBench-5*. VBench-5 is a compact paper-specific summary and is not the official VBench Quality, Semantic, or Total score. Paired analyses use prompt-seed pairs as statistical units.

Method naming and provenance. The immutable experiment archives predate the final terminology. Internal case identifiers containing `talh`, `c11`, or `stage2` denote InTraScale, ITU-only, and ITU, respectively. The code-and-data package contains an explicit mapping and retains the internal identifiers so that archived results remain traceable.

Evidence coverage. The package includes validated Wan50 and Distill4 quality, timing, component, and paired-statistics exports. It also includes the checksum-verified 50-sample raw $368 \times 640 \rightarrow 720 \times 1248$ ITU operator evaluation and a deterministic aggregation script. The local archive does not contain the complete custom ITU/TTD checkpoints or exact hardware/software version metadata. These missing items are not reconstructed from filenames or earlier prose.

Implementation and Training Details

In-Trajectory Upsampler

ITU maps a 16-channel clean video latent to the target spatial grid without changing the temporal dimension. A 3D input projection and four spatiotemporal residual blocks precede the spatial lifting operator; four further residual blocks and a 3D output projection reconstruct the target latent. The hidden width is 256. For the $3/2$ grid ratio, a $3 \times$ PixelShuffle is

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followed by anti-aliased stride-2 downsampling. For the approximately $2 \times$ ratio, a $2 \times$ PixelShuffle produces a 92×160 latent grid that is center-cropped to 90×156 .

ITU minimizes

$$\mathcal{L}_{\text{ITU}} = \mathcal{L}_{\text{rec}} + 0.2\mathcal{L}_{\text{lf}} + 0.1\mathcal{L}_{\text{temp}},$$

where the reconstruction term is Charbonnier loss with $\epsilon = 10^{-3}$, the low-frequency term preserves downsampled content, and the temporal term matches first-order latent differences.

Trajectory-Tail Distillation

TTD is a transition-local LoRA update. It targets the attention q, k, v, o projections and the two feed-forward projections, uses rank and training scale $r = \alpha = 32$, and keeps the base denoiser frozen. Its loss is

$$\mathcal{L}_{\text{TTD}} = \mathcal{L}_1 + 0.1\mathcal{L}_{\text{MSE}} + \lambda_{\text{tt}}\mathcal{L}_{\text{temp}}.$$

We use $\lambda_{\text{tt}} = 0.05$ for Wan50 step 40 and zero for Wan50 step 45 and Distill4 step 3. The inference strength is selected from $\{0.25, 0.5, 0.75, 1.0\}$ on held-out development prompts; 0.75 is then fixed for the test sets.

Setting	ITU	TTD
Optimizer	AdamW	AdamW
Maximum steps	50,000	10,000
Precision	bf16	bf16
Learning rate	1×10^{-4}	5×10^{-5}
Weight decay	0.01	0.01
Effective batch size	8	8
Gradient clipping	1.0	1.0
Training seed	1234	1234
EMA	0.9999	none
Validation fraction	0.05	0.02
Trainable parameters	ITU only	rank-32 LoRA only

Table 1: Final training settings encoded in the supplied configurations. ITU uses batch size 1 with distributed ranks/accumulation totaling an effective batch size of 8.

Evaluation Protocol

Generation. Wan50 uses Wan2.1 T2V-1.3B with 50 denoising evaluations and transition steps 40 or 45 for

InTraScale. Distill4 uses Wan2.1 T2V-14B StepDistill-CfgDistill with scheduler steps [1000, 750, 500, 250]; InTraScale-D4 performs three low-resolution evaluations and one target-resolution evaluation. Every output contains 81 frames at 720×1248 .

Prompts and randomness. Quality uses ten matched prompt–seed pairs in each suite. Wan50 uses seeds 9700–9709 and Distill4 uses seeds 9800–9809. The state-reconstruction and strength selection study uses eight disjoint prompts with seed base 16000. Prompt lists are included in the code-and-data package.

Warm latency. The model and checkpoints remain resident in one process. We run one untimed warm-up and then five matched generations. CUDA synchronization surrounds each measurement. Pipeline latency covers denoising, TTD, ITU, scheduler state reconstruction, VAE decoding, and output writing; it excludes runner construction and checkpoint loading. Exact GPU, driver, CUDA, and framework versions were not preserved in the local export and must be recovered from the measurement machine before final submission.

Statistics. For quality, we report prompt-paired means, nonparametric bootstrap 95% confidence intervals, win/loss counts, and two-sided sign tests. Timing variation is the sample standard deviation across the five warm repeats. These two forms of uncertainty are not interchanged.

Code, Data, and Media Inventory

The accompanying code-and-data ZIP contains the ITU model, TTD LoRA utilities, Wan50 and Distill4 bridges, data builders, training entry points, inference runners, VBench aggregation, paired-statistics scripts, final configurations, prompt lists, sanitized raw metric JSONs, and the compact CSV tables used here. The media ZIP contains two matched prompt–seed groups. Each group compares Trilinear, ITU-only, and InTraScale at 1248-by-720 resolution, 81 frames, and 16 frames per second. Legacy 640-by-368 files misleadingly named `Native-HR` are explicitly excluded.

Reproducibility boundary. Public base-model weights, large latent LMDs, and generated-video test sets are not duplicated. More importantly, the current local archive does not contain the final custom ITU/TTD weights. The package therefore supports independent audit of implementation, protocols, raw metric exports, and statistical claims, but full regeneration requires adding the final custom checkpoints.

Additional Wan50 Results

Method	LR evals	HR evals	VBench-5 $\uparrow$	Warm latency (s) $\downarrow$
Native-HR	0	50	0.82836	187.54 ± 0.26
Trilinear@40	40	10	0.77812	58.12 ± 0.20
InTraScale@40	40	10	0.80983	58.49 ± 0.16
Trilinear@45	45	5	0.76776	41.98 ± 0.19
RALU-style@45	45	5	0.80440	42.01 ± 0.19
InTraScale@45	45	5	0.80792	42.26 ± 0.20
Trilinear@48	48	2	0.76409	32.27 ± 0.18
ITU-0HR	50	0	0.79647	25.88 ± 0.11
ITU-1HR	50	1	0.80093	29.46 ± 0.22
ITU-2HR	50	2	0.80355	33.23 ± 0.10
ITU-5HR (LUVE-style)	50	5	0.80073	44.00 ± 0.27

Table 2: Wan50 warm-model operating points. Endpoint-first ITU baselines complete all 50 low-resolution evaluations before lifting and optional target-resolution refinement.

ITU Cross-Resolution Reconstruction

Input	Metric	Trilinear	ITU	Wins
480p	Latent $L_1 \downarrow$	0.19288	0.09913	49/50
480p	PSNR $\uparrow$	24.2927	29.9980	50/50
480p	SSIM $\uparrow$	0.71588	0.86961	49/50
480p	LPIPS $\downarrow$	0.23575	0.06294	50/50
480p	Temporal $L_1 \downarrow$	0.02121	0.01468	50/50
480p	HF-energy error $\downarrow$	0.00449	0.00161	48/50
368p	Latent $L_1 \downarrow$	0.24437	0.16284	50/50
368p	PSNR $\uparrow$	23.4238	24.1021	37/50
368p	SSIM $\uparrow$	0.67404	0.75920	48/50
368p	LPIPS $\downarrow$	0.33522	0.16403	50/50
368p	Temporal $L_1 \downarrow$	0.02388	0.02017	49/50
368p	HF-energy error $\downarrow$	0.00792	0.00233	50/50

Table 3: ITU operator evaluation on 50 clean-latent pairs per setting. “Wins” counts samples on which ITU is better. The raw 368p records and deterministic aggregation script accompany the submission.

TTD Endpoint Alignment

Transition	Base L_1	TTD L_1	Reduction	95% CI of improvement	Wins
Wan50 $s = 40$	0.03215	0.02385	25.82%	[0.00479, 0.01302]	10/10
Wan50 $s = 45$	0.02363	0.01866	21.03%	[0.00260, 0.00840]	10/10
Distill4 $s = 3$	0.04286	0.04070	5.03%	[0.00148, 0.00299]	10/10

Table 4: Prompt-paired endpoint alignment at the selected TTD strength $\lambda = 0.75$. Positive confidence intervals indicate lower endpoint L_1 after TTD.

Factorial Separation of ITU and TTD

Suite	Effect	Before	After	Δ VBench-5
Wan50@40	ITU within matched transition	0.77812	0.80896	+0.03085
Wan50@40	TTD within ITU	0.80896	0.80983	+0.00087
Wan50@45	ITU within matched transition	0.76776	0.80842	+0.04066
Wan50@45	TTD within ITU	0.80842	0.80792	−0.00050
Distill4@3	ITU within matched transition	0.81796	0.85670	+0.03874
Distill4@3	TTD within ITU	0.85670	0.85680	+0.00010

Table 5: Factorial quality effects. ITU supplies the dominant aggregate perceptual gain; TTD targets transition endpoint mismatch rather than acting as a general perceptual enhancer.

Additional Distill4 Results

Method	LR/HR	Subject	Background	Motion	Aesthetic	Imaging	VBench-5	Latency (s)
Native-HR4	0/4	0.97186	0.97557	0.99123	0.65203	0.71135	0.86041	42.49
Trilinear@3	3/1	0.95932	0.96937	0.98376	0.58954	0.58783	0.81796	18.33
InTraScale-D4@3	3/1	0.96226	0.97026	0.98815	0.64914	0.71418	0.85680	19.92
ITU-2HR (LUVE-style)	4/2	0.96238	0.97341	0.98683	0.65191	0.72450	0.85981	28.76
MrFlow-style	4/1	0.96379	0.97259	0.98796	0.65491	0.71591	0.85903	27.12

Table 6: Distill4 warm-model quality–efficiency. The last two methods are endpoint-first baselines and perform additional processing after completing the four-step low-resolution trajectory.

Baseline	Metric	Baseline–InTraScale	95% CI	Wins/Losses	p
Trilinear@3	VBench-5	−0.03884	[−0.04840, −0.02921]	0/10	0.00195
Trilinear@3	Temporal	−0.00264	[−0.00434, −0.00130]	1/9	0.02148
ITU-2HR	VBench-5	+0.00301	[0.00077, 0.00564]	7/3	0.34375
ITU-2HR	Temporal	−0.00333	[−0.00547, −0.00152]	1/9	0.02148
MrFlow-style	VBench-5	+0.00223	[−0.00277, 0.00752]	6/4	0.75391
MrFlow-style	Temporal	−0.00180	[−0.00289, −0.00075]	2/8	0.10938

Table 7: Distill4 prompt-paired statistics. Differences are baseline minus InTraScale; “Temporal” is temporal consistency, so higher is better. p is the two-sided sign-test value.

Scheduler-State Reconstruction and Strength Selection

Target-grid noise	TTD strength	VBench-5	Temporal consistency	Selected
Seed-derived Gaussian	0.25	0.83816	0.97476	no
Seed-derived Gaussian	0.50	0.83893	0.97468	no
Seed-derived Gaussian	0.75	0.84009	0.97465	yes
Seed-derived Gaussian	1.00	0.83933	0.97464	no
Resized LR flow	0.25	0.81467	0.95862	no
Resized LR flow	0.50	0.81488	0.95853	no
Resized LR flow	0.75	0.81526	0.95860	no
Resized LR flow	1.00	0.81486	0.95859	no

Table 8: Eight-prompt development study. Scheduler-consistent target-resolution noise improves both summary quality and temporal consistency across all tested TTD strengths.